

Science 24

UNIT D WORKBOOK



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Chapter 13: A Need For Safety

Define the following:

Statistics

Reaction Time

Distraction

Stimulant

Impaired

Depressant

BAC

Drink

Impaired driving

Trauma

Whiplash

Grief

Therapy

Maintenance

1. In Canada how much more likely will you die in a road collision than on an airplane? (pg. 234)
2. What type of injuries do you get in a car crash? (pg. 235)
3. Even if alert what are some factors that might affect your ability to stop? (pg. 237)
4. Why shouldn't you follow too closely? (pg. 237)
5. List 4 things that slow your reaction time. (pgs. 237, 240, 241)
6. Are cell phones a problem when driving? Why or why not? (pg. 237)

7. What should you do if you feel fatigued when driving? (pg. 240)

8. Why are stimulants dangerous to take when driving? (pg. 240)

9. Why does Alcohol slow reaction time? (pg. 240)

10. What is the legal blood alcohol limit? (pg. 241)

11. What factors affect your BAC? (pg. 241)

12. Put the following people in order from drunkest to Sober.

- a. A 300 lb male drinks 4 drinks in 2 hours.
- b. A 90 lb female who has 2 drinks in 1 hour.
- c. A 200 lb 55-year-old alcoholic who has 3 drinks in 2 hours
- d. A 90 lb female who has 2 drinks in 1 hour on a full stomach.

13. Use the following chart for the next question.

BAC Based on Number of Drinks												
Body Mass Max. (kg)	Female Number of Drinks						Male Number of Drinks					
	1	2	3	4	5	6	1	2	3	4	5	6
45	0.05	0.10	0.152	0.203	0.253	0.304	0.043	0.087	0.130	0.174	0.217	0.261
57	0.04	0.080	0.120	0.162	0.202	0.244	0.034	0.069	0.103	0.139	0.173	0.209
68	0.034	0.068	0.101	0.135	0.169	0.203	0.029	0.058	0.087	0.116	0.145	0.174
79	0.029	0.058	0.087	0.117	0.146	0.175	0.025	0.050	0.075	0.100	0.125	0.150
91	0.026	0.050	0.076	0.101	0.126	0.152	0.022	0.043	0.065	0.087	0.108	0.130
102	0.022	0.045	0.068	0.091	0.113	0.136	0.019	0.039	0.058	0.078	0.097	0.117
114	0.020	0.041	0.061	0.082	0.101	0.122	0.017	0.035	0.052	0.070	0.087	0.105

a. The body eliminates 0.015BAC units per hour. The legal limit is 0.05. If you drank 4 drinks in 1 hr how long will you have to wait before you can drive. (pg. 228)

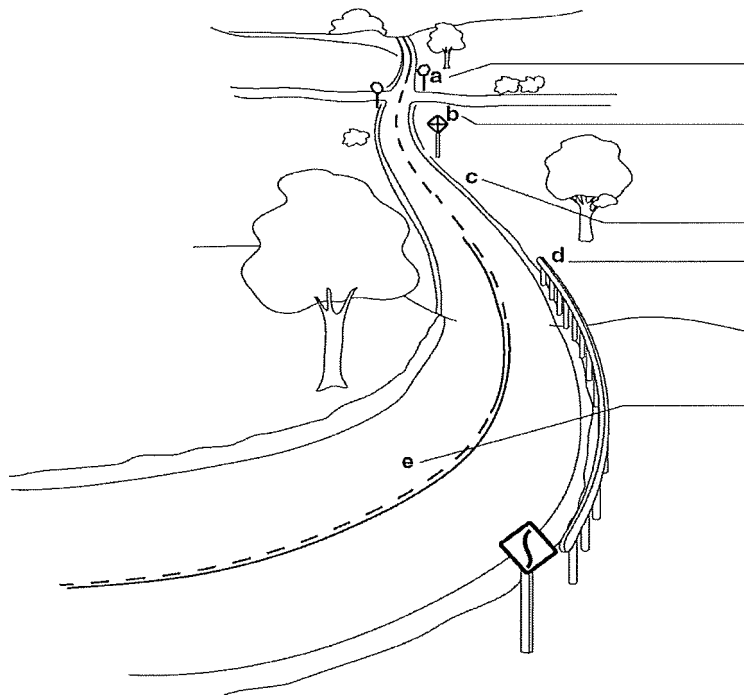
b. How long would it take a 68kg Female to sober up after 6 drinks in an hour?

14. What part of the car will prevent whiplash? (pg. 243)

15. List 4 safety features that your car has. (pg. 245)

16. Why should you perform regular maintenance on a vehicle? (pg. 245)

17. Label the safety features on this road. (pg. 247)



Chapter 14: The Nature of Speed

Define the following:

Distance

Time

Velocity

Distance/time graphs

Variables

Manipulated variable

Responding variable

Run

Rise

Slope

Braking distance

Stopping distance

Following distance

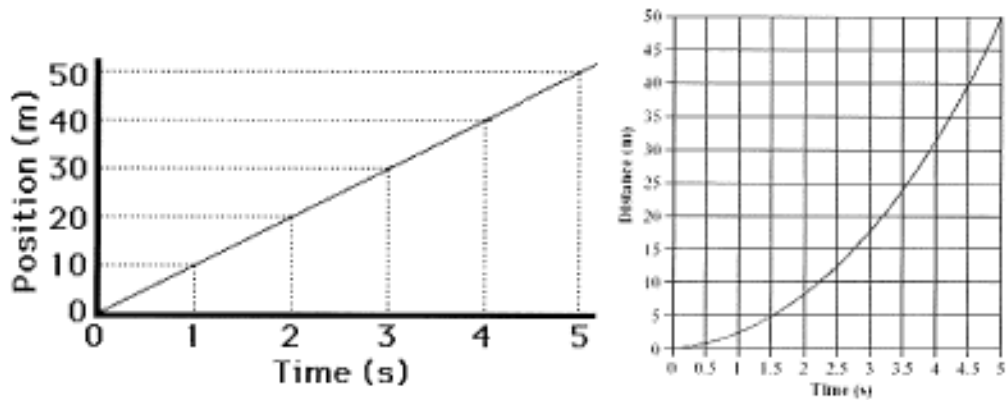
1. What is another word for velocity? (pg. 252)
2. What is the formula for calculating velocity? (pg. 254)
3. Rearrange the formula above to solve for distance. (pg. 254)
4. How does increasing the distance affect the time if the velocity stays the same? (pg. 254)
5. How would doubling the velocity while keeping distance the same affect time of travel? (pg. 254)
6. Cameron drives 300 km in 2 hours. How fast is he traveling? Show your work. (pg. 254)
7. Brandon is running to Macs to grab a slurpee at 10 minute break (600s). Round trip to Macs from MCHS is 1600m. Can Brandon make it in time if he runs 10m/s?

8. On a distance vs time graph on what axis does the time go? (pg. 256)

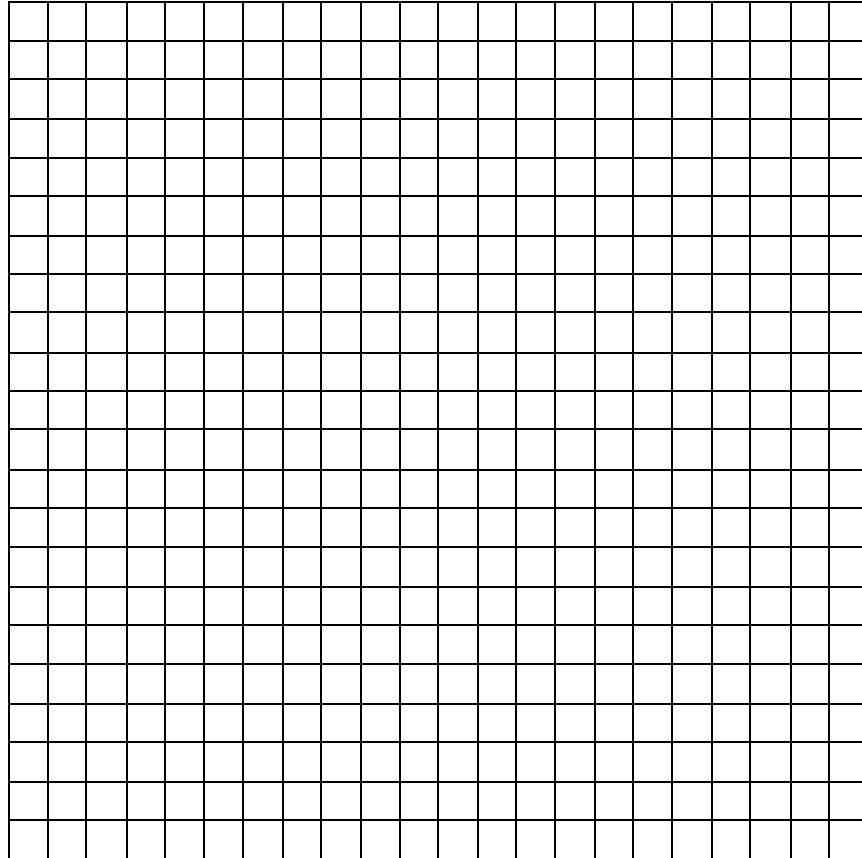
9. On a distance vs time graph what axis does the distance go? (pg. 256)

10. What type of line do we draw through the points on a distance vs time graph? (pg. 256)

11. **Circle** the following graph that show someone traveling at a constant speed? (pg. 257)

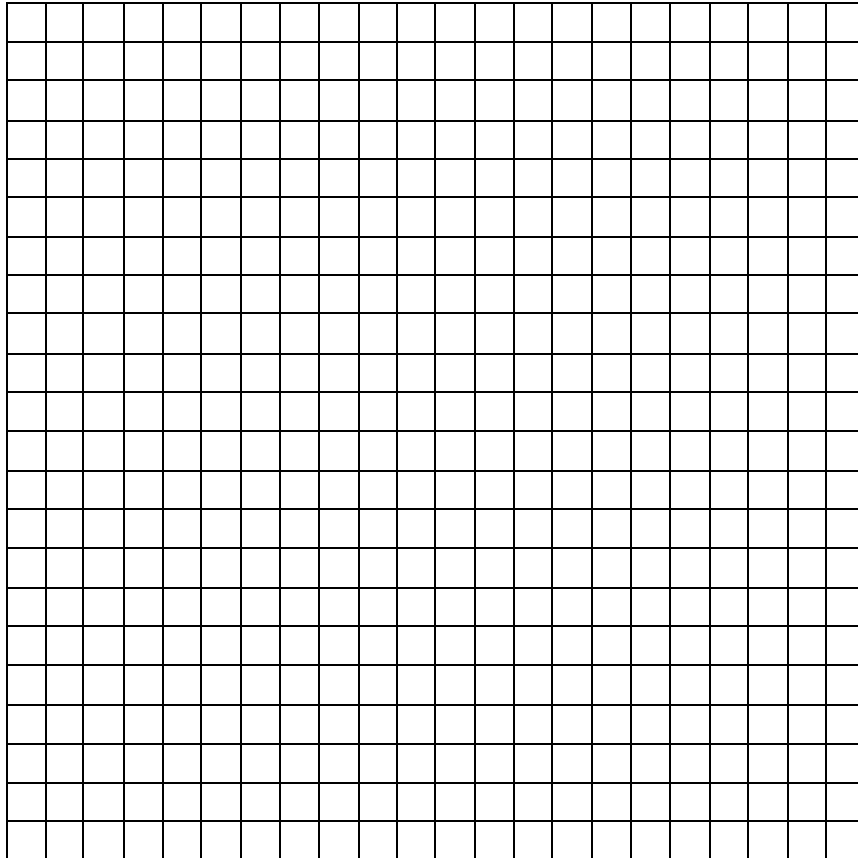


12. Draw a graph rough of the following situation. Tyler runs at a constant speed for 5 seconds. He then stops for 5 seconds and runs again.



13. Graph the following table of values on a graph. Be sure to create a title and to label all your axis.

Time	0 s	1 s	2 s	3 s	4 s	5 s
Distance	0 s	2 m	4 m	6 m	8 m	10 m

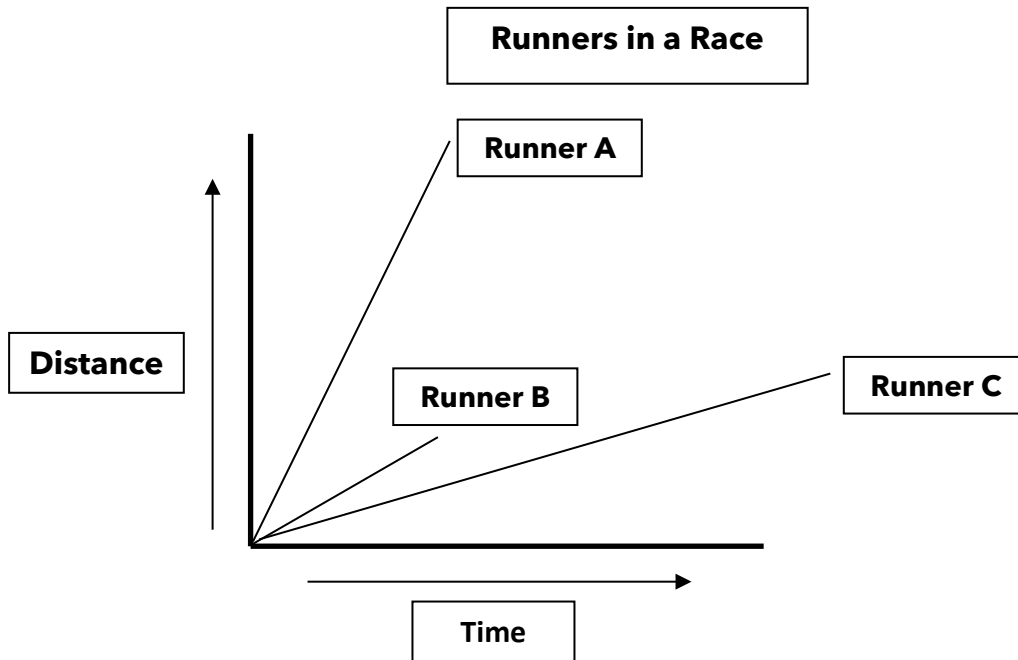


14. Calculate the slope of the object. (pg. 260)

15. What does the slope give us? (pg. 261)

16. What does the horizontal line a distance vs. time graph represent?

17. Use the following graph to answer the next 3 questions.



- a) Which runner was running the fastest?
- b) Which runner went the shortest distance?
- c) Which runner ran for the longest time?

18. Complete the following Chart.

Trip	Distance	Time	Speed
Edmonton to Red Deer		1.35 h	110 km/h
Jasper to Hinton		0.66 h	100 km/h
Morinville to Vegas	1966 km		120 km/h
Morinville to Drumheller	275 km	2 h	
Morinville to Jasper		3.11 h	100 km/h
Morinville to Edmonton	28 km	0.167 h (10 min)	

19. What is the difference between Braking distance and Stopping distance? (pg. 264)

20. When should you increase following distance? (pg. 265)

Chapter 15: The Physics of Collisions

Define the following:

Momentum

Force

Braking

Friction

Impulse

Law of Conservation of Momentum

Total momentum

1. When will an object that is not moving begin to move? (pg. 270)

2. When will an object that is moving stop? (pg. 270)

3. How do we calculate momentum? (pg. 271)

4. Fill in the following chart. (pg. 271)

Object	Mass	Velocity	Momentum
Compact Car	750kg	100km/h	
Adult female on a bike	70kg	30km/h	
Adult female	70kg	5km/h	
Transport truck	2500kg	50km/h	
Transport truck	2500kg	100km/h	

5. Give an example of an object with a very high momentum and one with a very low momentum. (pg. 271)

6. How is momentum affected in each situation. (pg. 271)

- a. Doubling velocity
- b. Doubling mass
- c. Doubling both velocity and mass

7. How do brakes work? (pg. 272)

8. Why don't brakes work as well in wet or icy conditions? (pg. 272)

9. How do we calculate impulse? (pg. 274)

10. Solve the following questions:

- a. A football player collides with another player. The force of the collision is 100N. The collision lasts 2 seconds. What is the impulse?
- b. A baseball player feels a frictional force of 400N as she slides into home. What is the impulse if the slide lasts 0.5 seconds?

11. How does increasing your stop time protect you from injury? (pg. 274)

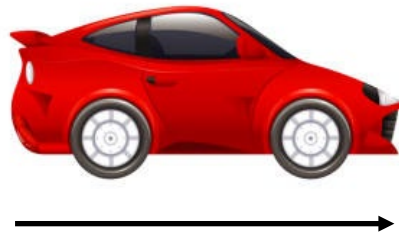
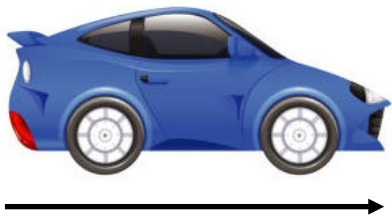
12. Is the force needed to stop a car the same whether you take a long time or a short time to stop? Explain.

13. In a collision that stops a vehicle, how does the original velocity relate to the force experienced by vehicle and passenger? In other words, if a car hits a brick wall how does the speed at which it hits the wall affect the force felt by the passenger? (pg. 277)

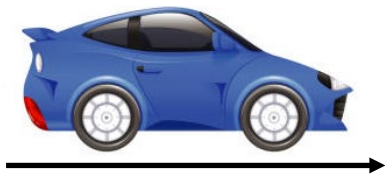
14. Use the following diagrams and determine what happens to momentum. (pg. 281)
a. Two cars of the same size and traveling at the same speed hit in a head on collision.



b. Draw the arrows and explain what would happen to momentum after collision.
Two cars traveling in the same direction hit.



c. What would happen in a head on collision between these two vehicles?
Assuming both vehicles were traveling at the same speed.



Chapter 16: The Technology of Safety

Define the following:

Crash test dummies

Crumple zones

Restraining features

Operational features

Structural features

Road safety features

Crash cushion

Guardrails

2-point seat belts

Lap belts

1. What do safety features do on a car? (pg. 288)

2. Why do vehicles have crumple zones? (pg. 288)

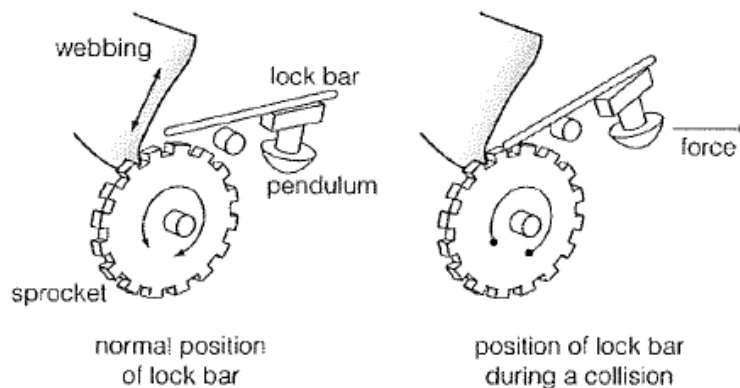
3. List 4 examples of a restraining feature. (pg. 289)

4. Why are your rear and side mirrors an example of an operational feature? (pg. 289)

5. What are the two types of structural features? List an example of each. (pg. 289)

6. Why are the rigid features of a car surrounding the cabin of the vehicle? (pg. 289)

7. What is a major cause of injury in a car with no crumple zones? Why? (pg. 290)
8. List 4 road features that were designed to reduce the force of impact. (pg. 291)
9. What type of injury would happen if 2-point belts were in the front seat? (pg. 292)
10. Engineers have not found a way to restrain the head during a collision.
What safety features are designed to protect the head? (pg. 292)
11. Why does the webbing of a seat belt stretch? (pg. 293)
12. Explain how this structure works. (pg. 293)



13. Circle the right answer:

Babies heads are (**larger/ smaller**) in proportion to their body. (pg. 296)

14. Why should car seats be facing the rear of the vehicle? (pg. 297)

15. What type of gas does an air bag fill up with? (pg. 298)

16. What was the problem with the first-generation air bag? (pg. 298)

17. List 2 features of the third-generation air bags. (pg. 301)